

Classification II

Linear Discriminant Analysis (LDA)

- ▶ Classify an observation into one of K classes
- ▶ Data: $\mathbf{X}$ = continuous inputs
 Y = categorical output
- ▶ $\pi_k = P(Y = k)$ prior probability that a randomly chosen observation comes from class k
- ▶ $f_k(x)$ = probability density function for $(X|Y = k)$

Bayes theorem

- ▶ Starting with the definition of conditional probability



$$P(Y = k|X = x) \equiv \frac{P(X = k, X = x)}{P(X = x)}$$

- ▶ For discrete X -variables:

$$P(Y = k|X = x) = \frac{P(X = x|Y = k)P(Y = k)}{P(X = x)}$$



$$\begin{aligned} P(X = x) &= \sum_{k=1}^K P(X = x, Y = k) \\ &= \sum_{k=1}^K P(X = x|Y = k)P(Y = k) \end{aligned}$$

- ▶ For discrete X -variables:

$$P(Y = k|X = x) = \frac{P(X = x|Y = k)P(Y = k)}{\sum_{k=1}^K P(X = x|Y = k)P(Y = k)}$$

- ▶ Possible to show that for continuous X :

$$p_k(x) \equiv P(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{k=1}^K \pi_k f_k(x)}$$

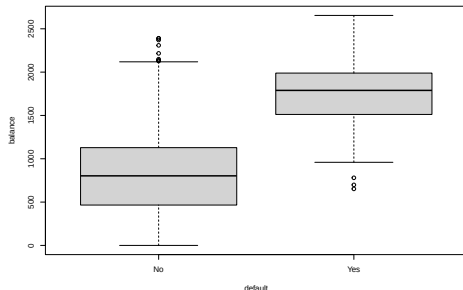
- ▶ $p_k(x_i)$ is the posterior probability that observation i belongs to class k , given that $X_i = x_i$.
- ▶ Bayes classifier: Predict, “allocate”, observation nr i to the class k with highest $p_k(x_i)$.

Default data

```
require(ISLR)
```

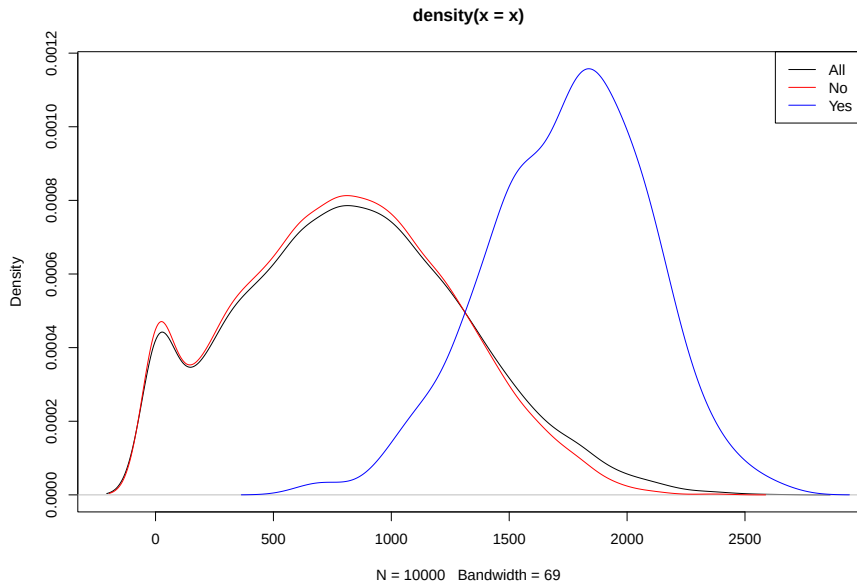
```
## Loading required package: ISLR
```

```
plot(balance~default,data=Default)
```



Default observations have a higher average account balance

Default data



Default data

```
by(Default$balance,Default$default,mean)
```

```
## Default$default: No
```

```
## [1] 803.9438
```

```
## -----
```

```
## Default$default: Yes
```

```
## [1] 1747.822
```

The case with one X -variable

- ▶ $p = 1$ (one X -variable) and $K = 2$ (two classes)
- ▶ We assume $(X|Y = k) \sim N(\mu_k, \sigma^2)$, $k = 1, 2$
- ▶ Prior probabilities $P(Y = k) = \pi_k$, $k = 1, 2$
- ▶ Densities: $f_k(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu_k)^2}{2\sigma^2}}$, $k = 1, 2$
- ▶ Reminder:

$$p_k(x) \equiv P(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{k=1}^K \pi_k f_k(x)}$$



$$p_k(x) = \frac{e^{-\frac{(x-\mu_k)^2}{2\sigma^2}}}{e^{-\frac{(x-\mu_1)^2}{2\sigma^2}} + e^{-\frac{(x-\mu_2)^2}{2\sigma^2}}}$$

- ▶ Allocate i to class 1 if $p_1(x_i) > p_2(x_i)$

Solving the one-variable case



$$p_1(x) > p_2(x)$$

$$\iff$$

$$e^{-\frac{(x-\mu_1)^2}{2\sigma^2}} > e^{-\frac{(x-\mu_2)^2}{2\sigma^2}}$$

$$\iff$$

$$(x - \mu_1)^2 < (x - \mu_2)^2$$

- ▶ Allocate i to class 1 if distance to μ_1 shorter than distance to μ_2 . Make sense?

Find Bayes classifier



$$p_k(x) = \frac{\pi_k f_k(x)}{\underbrace{\sum_{k=1}^K \pi_k f_k(x)}_{\text{same for all } k}}$$

- Largest $p_k(x) \iff$ Largest $\pi_k f_k(x) \iff$ Largest $(\ln \pi_k + \ln f_k(x))$



$$\begin{aligned} \ln \pi_k + \ln f_k(x) &= \ln \pi_k + \ln \left[\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu_k)^2}{2\sigma^2}} \right] \\ &= \ln \pi_k + \frac{1}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(x^2 - 2\mu_k x + \mu_k^2) \\ &= \ln \pi_k + \underbrace{\frac{1}{2} \ln(2\pi\sigma^2) - \frac{x^2}{2\sigma^2}}_{\text{constant over } k \text{ for given } x} + \frac{\mu_k x}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} \end{aligned}$$

- ▶ We use the *linear* discriminant function

$$\delta_k(x) = \ln \pi_k + \frac{\mu_k x}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2}$$

- ▶ i.e. allocate i to the class with largest $\delta_k(x_i)$
- ▶ Becomes *quadratic* function if the variance is not equal for the two classes. Quadratic Discriminant Analysis (QDA)

Empirical implementation - unknown parameters



$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$$



$$\hat{\sigma}_k^2 = \frac{1}{n - K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$



$$\hat{\pi}_k = \frac{n_k}{n} = \frac{\text{number of obs in class } k}{\text{number of obs}}$$

Implementing on Default data

```
x=Default$balance
cl=Default$default
nk=table(cl)
n=length(cl)
pi=nk/n
mu=as.matrix(by(x,cl,mean))
s2=(1/(n-2))*(sum((x[cl=="No"]-mu[1])^2)+sum((x[cl=="Yes"]-mu[2])^2))
mu
```

```
##           [,1]
## No      803.9438
## Yes    1747.8217
s2
```

```
## [1] 205318.6
pi
```

```
## cl
##      No      Yes
## 0.9667 0.0333
```

```

delta=function(x,k,mu,s2,pi) mu[k]*x/s2-mu[k]^2/(2*s2)+log(pi[k])
delta1=delta(x,1,mu,s2,pi)
delta2=delta(x,2,mu,s2,pi)
pred=factor(rep("No",n),levels=c("No","Yes"))
pred[pred=="No"]="Yes"
head(cbind(cl,pred))

```

```

##      cl pred
## [1,]  1    1
## [2,]  1    1
## [3,]  1    1
## [4,]  1    1
## [5,]  1    1
## [6,]  1    1

```

```
table(pred,cl)
```

```

##      cl
## pred   No  Yes
##   No 9643 257
##   Yes  24  76

```

$p > 1$, more than one X -variable

- ▶ Multivariate normally distributed $\mathbf{X} \in N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$



$$f_k(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{p}{2}} |\boldsymbol{\Sigma}|} e^{\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

- ▶ Similar to the univariate case it is possible to show that



$$\delta_k(x) = \ln \pi_k + \mathbf{x}' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k - \frac{1}{2} \boldsymbol{\mu}_k' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k$$

- ▶ is the discriminant function in the p -variate case if $\mathbf{X}$ have the same variance-covariance matrix for all groups.

Comparison with logistic regression

- ▶ When classes are well separated, parameter estimates in logistic regression are unstable. Not a problem in LDA.
- ▶ Small n and approximately multinormally distributed $\mathbf{X}$, LDA more stable than logistic regression.
- ▶ A natural approach when Y has many classes (albeit an extension to logistic regression is available)

Confusion matrix

	<i>Predicted class</i>	
<i>True class</i>	True Neg (TN)	False Pos (FP)
	False Neg (FN)	True Pos (TP)

```
##      pred
## cl      No  Yes
##  No  9643   24
##  Yes  257   76
```

$$\text{True positive rate} = \frac{\text{TP}}{\text{FN} + \text{TP}} = \frac{76}{257 + 76} = 0.23$$

$$\text{False positive rate} = \frac{\text{FP}}{\text{TN} + \text{FP}} = \frac{24}{9643 + 24} = 0.0025$$

Threshold - value

- ▶ We choose the class maximizing the posterior probability of belonging to a class. If there are only two classes, a natural threshold is 0.5; but we could use others.
- ▶ Default example: Classify an observation i as default if $p_2(x_i) > 0.5$; or if it is > 0.2 . Try several alternatives to evaluate a model, e.g. LDA or logistic regression.

ROC-curve

Plot True Positive Rate against False Positive Rate for many different thresholds.

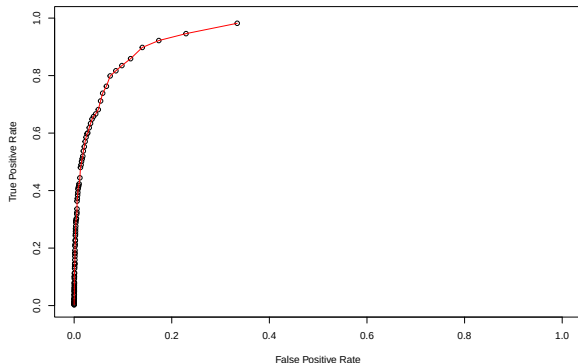
```
require(MASS)
```

```
## Loading required package: MASS
```

```
lda1=lda(cl~x)  
pr=predict(lda1)$posterior  
head(pr)
```

```
##           No           Yes  
## 1 0.9972130 0.002786981  
## 2 0.9958358 0.004164240  
## 3 0.9865931 0.013406929  
## 4 0.9988882 0.001111757  
## 5 0.9963955 0.003604464  
## 6 0.9933487 0.006651334
```

ROC-curve



AUC = Area Under Curve gives a summary measure of the model's performance.

ROC-curve

```
thrang=seq(0.01,0.95,0.01)
nth=length(thrang)
roc=data.frame(FPrate=0,TPrate=0)
ii=1
for(th in thrang)
{
  pred=pr[,2]>th
  cm=table(cl,pred)
  fprate=cm[1,2]/sum(cl=="No")
  tprate=cm[2,2]/sum(cl=="Yes")
  roc[ii,]=cbind(fprate,tprate)
  ii=ii+1
}
plot(roc,xlim=c(0,1),ylim=c(0,1))
lines(roc,col="red")
lines(roc[1,],c(1,1),col="red")
```

K-nearest neighbors classifier

- ▶ x_0 is a *prediction point* (a test observation)
- 1. Find the K training-points that are closest to x_0 , call this set $\mathbb{N}_0$.
- 2. $P(Y = k|X = x_0) = \frac{1}{K} \sum_{x_i \in \mathbb{N}_0} I(y_i = k)$ where I is the indicator function.
- ▶ In words: Allocate the new observation to the class for which a plurality of the K closest neighbors belong.